U-W-00138a(DSA-DM)
24 January 1967
Interim Revision of
Fed. Spec. U-W-138
12 May 1947

INTERIM FEDERAL SPECIFICATION

WAX, DENTAL (BOXING)

This interim Federal specification was developed by the Defense Personnel Support Center, 2800 South 20th Street, Philadelphia, Pennsylvania 19101, based upon currently available technical information. It is recommended that Federal agencies use it in procurement and forward recommendations for changes to the preparing activity at the address shown above. The General Services Administration has authorized the use of this interim Federal specification as a valid exception to Federal Specification U-W-138.

1. SCOPE

1.1 Scope. This specification covers boxing wax suitable for use as a frame to contain gypsum material in dental prosthetic procedures.

2. APPLICABLE DOCUMENTS

2.1 Specifications and standards. The following specifications and standards, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

PPP-B-585	Boxes, Wood, Wirebound.
PPP-B-601	Boxes, Wood, Cleated-Plywood.
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	Box, Fiberboard.
PPP-B-676	Box, Setup.

Military

MIL-L-10547	Liners, Case, and Sheet, Overwrap; Water-Vaporproof, or Waterproof, Flexible.

FSC 6520

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STANDARDS

Federal

FED-STD-123 Marking for Domestic Shipment (Civilian Agencies).

Military

MIL-STD-105 Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 Marking for Shipment and Storage.

(Activities outside the Federal Government may obtain copies of Federal Specifications and Standards as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C., 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, Seattle, and Washington, D.C.

(Federal Government activities may obtain copies of Federal Specifications and Standards and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

(Copies of Military specifications, standards, and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS

ASTM E28-58T Method of Test for Softening Point by Ring and Ball Apparatus (Tentative).

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pa. 19103).

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3. REQUIREMENTS

3.1 Material. Boxing wax shall be a uniform mixture of waxes, and other ingredients, such as resins, combined in proportions that will result in a material possessing the physical properties hereinafter specified.

3.2 Description. The boxing wax shall be suitable for use as a frame to contain gypsum inpression material in dental prosthetic procedures. The wax shall be tacky, dark red in color, and shall cohere with finger pressure at room temperature. The wax shall be furnished in smooth sheets 12 ± 1/8 inches long, 1 1/2 ± 1/32 inches wide, and 0.040 ± 0.003 inch thick.

3.3 Physical properties.

3.3.1 Softening point. The wax shall have a softening point between 150.0 degrees Fahrenheit (°F.) and 172.0° F. when tested as specified in 4.4.1.

3.3.2 Breaking point. The wax shall not fracture when bent double at a temperature of 70.0° ± 2.0° F. when tested as specified in 4.4.2.

3.4 Working properties. The wax shall have the following characteristic working properties:

a. Shall be pliable at 70.0° ± 2.0° F., yet stiff enough to retain position and shape at 95.8 ± 2.0° F.

b. Shall be capable of sealing easily to a plaster model with hand manipulation at 70.0° ± 2.0° F.

3.5 Weight. The wax shall be furnished in unit boxes containing a minimum of one pound of material.

3.6 Lot or batch number. Each unit package shall be marked with a lot or batch number and date of manufacture.

3.7 Stability. The wax shall conform to the requirements of the specification for not less than two years from date of delivery to the Government, when stored in a cool, dry place (50° to 80° F.).

3.8 Delivery schedule. Not more than two months shall have elapsed from date of manufacture of the wax to date of delivery to the Government.

3.9 Workmanship. Workmanship shall be first class throughout. The boxing wax shall be free from defects which detract from its appearance or may impair its serviceability.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsiblity for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties).

4.1.2 Certificates of quality. When available, certificates of quality, supplied by the manufacturer of the component or material, listing the specified test method and test results obtained, may be furnished in lieu of actual performance of such testing by the contractor.

4.2 Sampling.

4.2.1 For examination. Sampling for examination shall be conducted in accordance with MIL-STD-105.

4.2.2 For tests. Sampling for tests shall be conducted as indicated in table I. The unit of product for inspection purposes shall be one package of boxing wax.

Table I. Sampling For Tests

1	2	3	4	5	6
Characteristic	Reqmt.	Test procedure	Unit of product to be tested per lot or batch number	Number of determinations per unit of product	Total number of determinations per lot or batch number
Softening point	3.3.1	4.4.1	1	3	3
Breaking point	3.3.2	4.4.2	2	3	6
Working properties	3.4	4.4.3	2	3	6
Weight	3.5	S.M.I.*	3	1	3

* SMI - Standard Measuring Instruments

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4.2.3 Where feasible, the same sample unit shall be used for the determination of two or more test characteristics.

4.3 Examination. The boxing wax shall be examined to determine compliance with specification requirements.

4.4 Tests.

4.4.1 Softening point. The softening point of the wax shall be determined by method ASTM Designation: E 28-58T.

4.4.2 Breaking point. A 6 ± 1/8 inch sample of boxing wax at 70.8 ± 2.0° F shall be bent in 10 seconds at a uniform rate until it is bent double with the two halves of the specimen in contact for their entire length.

4.4.3 Working properties. The boxing wax shall be given a practical performance test to determine compliance with the specified property requirements cited in 3.4.

5. PREPARATION FOR DELIVERY

5.1 Packaging.

5.1.1 Unit of issue. One box containing one pound of wax, as specified, constitutes one unit of issue.

5.1.2 Packaging quantities. The number of units of issue indicated in table II shall be packaged in each unit and exterior container, as applicable, for the required level of protection specified in the procurement document.

Table II. Packaging Quantities

Unit package	Intermediate package	Exterior container
1 unit	Not reqd.	48 units

5.1.2.1 Packing variation permitted. If the required number of units to be shipped is less than the number of units specified to be overpacked in an exterior container, such units shall be packed in an exterior container of suitable size and design, acceptable to a common carrier, which shall insure safe delivery to destination.

5.1.3 Level A.

5.1.3.1 Unit package. The layers of wax sheets shall be separated with silicon paper and packaged in a carton of appropriate size constructed in accordance with PPP-B-676, except that commercial colors are acceptable. Closure of carton shall be adequate to prevent spilling of contents under normal handling.

5.1.4 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

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5.2 Packing.

5.2.1 Level A.

5.2.1.1 Exterior container. Exterior container shall be designed for type 2 load, and constructed in accordance with PPP-B-585, style 3, for class 3 use; PPP-B-601, table II, using type III, class 1 plywood; PPP-B-621, table III; or PPP-B-636, type CF, class weather-resistant.

5.2.1.2 Case liner. Each level A wood box shall be lined with a waterproof case liner conforming to MIL-L-10547. Closure and sealing shall conform to applicable paragraphs of appendix thereto. Case liner shall not be required for fiberboard boxes.

5.2.1.3 Closure. Closure of wood boxes shall be in accordance with appendix of applicable box specification. Closure of each fiberboard box shall be as specified in the appendix, and waterproofing shall conform with 30.4, of PPP-B-636.

5.2.1.4 Strapping. Strapping, when required, shall be in accordance with appendix of applicable box specification.

NOTE: Strapping shall not be required for shipments forwarded to a receiving activity within the continental limits of the United States for storage and redistribution.

5.2.2 Level B.

5.2.2.1 Exterior container. Exterior container shall conform to PPP-B-601, table I, using type III, class 1 plywood; or shall be designed for type 2 load, and constructed in accordance with PPP-B-585, style 3, for class 2 use; PPP-B-621, table I; or PPP-B-636, type CF, class domestic and method I closure shall be utilized.

5.2.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.3 Marking.

5.3.1 Civil agencies. Shipments shall be marked in accordance with FED-STD-123.

5.3.2 Military agencies.

5.3.2.1 Unit package. Each unit package shall be marked in accordance with MIL-STD-129. When labels are utilized, waterproofing shall be required only when applicable carton is fabricated of water-resistant material. Contract or purchase order number and name of contractor shall be shown.

5.3.2.2 Exterior container. Exterior container shall be marked in accordance with MIL-STD-129.

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6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Number and date of this specification.

(b) Selection of applicable levels of packaging and packing (see 5.1 and 5.2)

6.2 This specification does not cover all types, classes, grades, or sizes of the commodity indicated by the title of this specification, or those which are commercially available, but is intended to cover the types which are normally procured to meet requirements.

Custodians:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
Defense Supply Agency - DM

Review activities:
Army - MD
Navy - MS
Air Force - 03

Review information is current as of the date of this document. For future coordination of changes to this document, draft circulation should be based on the information in the current DODISS.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

U-W-00138a(DSA-DM)

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64, WHICH MAY BE USED.

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